$= \sqrt[3]{2^4 . 2^{(-1)}} = \sqrt[3]{2^{(4-1)}} = \sqrt[3]{2^3} = (2^3)^{\frac{1}{3}}$

$= 2^{\left(3\times\frac{1}{3}\right)} = 2.$

26. $\sqrt{2\sqrt{2\sqrt{2\sqrt{2\sqrt{2}}}}} = \sqrt{2\sqrt{2\sqrt{2\sqrt{2.2^{\frac{1}{2}}}}}} = \sqrt{2\sqrt{2\sqrt{2\sqrt{2^{\left(1+\frac{1}{2}\right)}}}}}$

$= \sqrt{2\sqrt{2\sqrt{2\sqrt{2^{\frac{3}{2}}}}}} = \sqrt{2\sqrt{2\sqrt{2.\left(2^{\frac{3}{2}}\right)^{\frac{1}{2}}}}}$

$= \sqrt{2\sqrt{2\sqrt{2.2^{\left(\frac{3}{2}\times\frac{1}{2}\right)}}}} = \sqrt{2\sqrt{2\sqrt{2.2^{\frac{3}{4}}}}}$

$= \sqrt{2\sqrt{2\sqrt{2^{\left(1+\frac{3}{4}\right)}}}} = \sqrt{2\sqrt{2\sqrt{2^{\frac{7}{4}}}}}$

$= \sqrt{2\sqrt{2.\left(2^{\frac{7}{4}}\right)^{\frac{1}{2}}}} = \sqrt{2\sqrt{2.2^{\left(\frac{7}{4}\times\frac{1}{2}\right)}}}$

$= \sqrt{2\sqrt{2.2^{\frac{7}{8}}}} = \sqrt{2\sqrt{2^{\left(1+\frac{7}{8}\right)}}} = \sqrt{2\sqrt{2^{\frac{15}{8}}}}$

$= \sqrt{2\left(2^{\frac{15}{8}}\right)^{\frac{1}{2}}} = \sqrt{2.2^{\frac{15}{16}}} = \sqrt{2^{\left(1+\frac{15}{16}\right)}}$

$= \sqrt{2^{\frac{31}{16}}} = \left(2^{\frac{31}{16}}\right)^{\frac{1}{2}} = 2^{\left(\frac{31}{16}\times\frac{1}{2}\right)} = 2^{\frac{31}{32}}.$

27. $(0.03125)^{-\frac{2}{5}} = \left[(0.5)^5\right]^{-\frac{2}{5}} = 0.5^{\left[5\times\left(-\frac{2}{5}\right)\right]} = (0.5)^{-2}$

$= \frac{1}{(0.5)^2} = \frac{1}{0.25} = 4.$

28. $\left(\frac{1}{2}\right)^{-\frac{1}{2}} = (2)^{\frac{1}{2}} = \sqrt{2}.$

29. $\left[\left(\sqrt[5]{x^{-\frac{3}{5}}}\right)^{-\frac{5}{3}}\right]^5 = \left[\left\{\left(x^{-\frac{3}{5}}\right)^{\frac{1}{5}}\right\}^{-\frac{5}{3}}\right]^5 = \left[\left(x^{\left\{\left(-\frac{3}{5}\right)\times\frac{1}{5}\right\}}\right)^{-\frac{5}{3}}\right]^5$

$= \left[\left(x^{-\frac{3}{25}}\right)^{-\frac{5}{3}}\right]^5 = \left[x^{\left\{\left(-\frac{3}{25}\right)\times\left(-\frac{5}{3}\right)\right\}}\right]^5$

$= \left(x^{\frac{1}{5}}\right)^5 = x^{\left(\frac{1}{5}\times 5\right)} = x.$

30. Let $\frac{x^{\frac{2}{3}}}{42} = \frac{5}{x^{\frac{1}{3}}}.$

Then, $x^{\frac{2}{3}} . x^{\frac{1}{3}} = 42\times 5 = 210 \Rightarrow x^{\left(\frac{2}{3}+\frac{1}{3}\right)} = 210 \Rightarrow x = 210.$

31. $5^{\frac{1}{4}}\times(125)^{0.25} = 5^{0.25}\times(5^3)^{0.25} = 5^{0.25}\times 5^{(3\times 0.25)}$

$= 5^{0.25}\times 5^{0.75} = 5^{(0.25+0.75)} = 5^1 = 5.$

32. $\frac{1}{(216)^{-\frac{2}{3}}} + \frac{1}{(256)^{-\frac{3}{4}}} + \frac{1}{(32)^{-\frac{1}{5}}}$

$= \frac{1}{(6^3)^{-\frac{2}{3}}} + \frac{1}{(4^4)^{\left(-\frac{3}{4}\right)}} + \frac{1}{(2^5)^{-\frac{1}{5}}}$

$= \frac{1}{6^{3\times\frac{(-2)}{3}}} + \frac{1}{4^{4\times\frac{(-3)}{4}}} + \frac{1}{2^{5\times\frac{(-1)}{5}}} = \frac{1}{6^{-2}} + \frac{1}{4^{-3}} + \frac{1}{2^{-1}}$

$= (6^2 + 4^3 + 2^1) = (36 + 64 + 2) = 102.$

33. $(2.4\times 10^3)\div(8\times 10^{-2}) = \frac{2.4\times 10^3}{8\times 10^{-2}} = \frac{24\times 10^2}{8\times 10^{-2}} = (3\times 10^4).$

34. $\left(\frac{1}{216}\right)^{-\frac{2}{3}} \div \left(\frac{1}{27}\right)^{-\frac{4}{3}} = (216)^{\frac{2}{3}} \div (27)^{\frac{4}{3}} = \frac{(216)^{\frac{2}{3}}}{(27)^{\frac{4}{3}}} = \frac{(6^3)^{\frac{2}{3}}}{(3^3)^{\frac{4}{3}}}$

$= \frac{6^{\left(3\times\frac{2}{3}\right)}}{3^{\left(3\times\frac{4}{3}\right)}} = \frac{6^2}{3^4} = \frac{36}{81} = \frac{4}{9}.$

35. $(48)^{-\frac{2}{7}}\times(16)^{-\frac{5}{7}}\times(3)^{-\frac{5}{7}} = (16\times 3)^{-\frac{2}{7}}\times(16)^{-\frac{5}{7}}\times(3)^{-\frac{5}{7}}$

$= (16)^{-\frac{2}{7}}\times(3)^{-\frac{2}{7}}\times(16)^{-\frac{5}{7}}\times(3)^{-\frac{5}{7}}$

$= (16)^{\left(-\frac{2}{7}-\frac{5}{7}\right)}\times(3)^{\left(-\frac{2}{7}-\frac{5}{7}\right)}$

$= (16)^{\left(-\frac{7}{7}\right)}\times(3)^{\left(-\frac{7}{7}\right)} = (16)^{-1}\times(3)^{-1}$

$= \frac{1}{16}\times\frac{1}{3} = \frac{1}{48}.$

36. $10^{-8x} = (10^x)^{-8} = \left(\frac{1}{2}\right)^{-8} = 2^8 = 256.$

37. $\left(\frac{3}{5}\right)^3\left(\frac{3}{5}\right)^{-6} = \left(\frac{3}{5}\right)^{2x-1} \Rightarrow \left(\frac{3}{5}\right)^{(3-6)} = \left(\frac{3}{5}\right)^{2x-1}$

$\Rightarrow \left(\frac{3}{5}\right)^{-3} = \left(\frac{3}{5}\right)^{2x-1}$

$\Rightarrow$ $2x - 1 = -3$ $\Rightarrow$ $2x = -2$ $\Rightarrow$ $x = -1.$

38. $49 \times 49 \times 49 \times 49 = 7^2 \times 7^2 \times 7^2 \times 7^2 = 7^{(2+2+2+2)} = 7^8.$

$\therefore$ Required number = 8.

39. $8^{-25} - 8^{-26} = \left(\frac{1}{8^{25}} - \frac{1}{8^{26}}\right) = \frac{(8-1)}{8^{26}} = 7\times 8^{-26}.$

40. $(64)^{-\frac{1}{2}} - (-32)^{-\frac{4}{5}} = (8^2)^{-\frac{1}{2}} - [(-2)^5]^{-\frac{4}{5}}$

$= 8^{\left[2\times\left(\frac{-1}{2}\right)\right]} - (-2)^{\left[5\times\left(\frac{-4}{5}\right)\right]}$

$= 8^{-1} - (-2)^{-4} = \frac{1}{8} - \frac{1}{(-2)^4}$

$= \left(\frac{1}{8} - \frac{1}{16}\right) = \frac{1}{16}.$